CSE411: Real-Time and Embedded Systems Design
CSE323: Advanced Embedded Systems Design
Spring 2026
Project Description
Smart Parking Garage Gate System

1. Project Overview

This project implements a Smart Parking Garage Gate System using the Tiva-C TM4C123GH6PM microcontroller running FreeRTOS. The system simulates an automated parking gate that can be controlled from both a driver's panel and a security panel. It includes safety features such as obstacle detection and motion limits.

2. Project Objectives

- Apply real-time operating system concepts using FreeRTOS
- Design a multitasking embedded system
- Implement safe gate control using limit buttons
- Use RTOS mechanisms: tasks, queues, semaphores, and mutexes
- Develop a finite state machine for system behavior

3. Hardware Components

Component	Description
Tiva-C TM4C123GH6PM	Main microcontroller
Green LED	Indicates gate moving UP (opening)
Red LED	Indicates gate moving DOWN (closing)
Push Button - Driver OPEN	Driver requests gate to open
Push Button - Driver CLOSE	Driver requests gate to close
Push Button - Security OPEN	Security requests gate to open
Push Button - Security CLOSE	Security requests gate to close
Push Button - Open Limit	Simulates gate reaching fully open position
Push Button - Closed Limit	Simulates gate reaching fully closed position
Push Button - Obstacle	Simulates obstacle detection
Status LEDs (Optional)	Display current system state

How Simulation Works

Since this is a simulation without a real motor, the limit buttons simulate the physical gate position:

Action	Simulation
Gate opening	Green LED turns ON. Press Open Limit button when you want to simulate gate reaching fully open.
Gate closing	Red LED turns ON. Press Closed Limit button when you want to simulate gate reaching fully closed.
Obstacle detected	While gate is closing (Red LED ON), press Obstacle button to simulate an obstacle.

4. System Functional Requirements

1. Manual Mode

When the OPEN or CLOSE button is held, the gate moves only while the button remains pressed. Releasing the button stops the gate.

2. One-Touch Auto Mode

A brief press of the button causes the gate to move fully in that direction until the corresponding limit button is pressed (simulating gate reaching the limit).

3. Obstacle Protection

If an obstacle is detected while the gate is auto-closing:

- Gate stops immediately
- Gate reverses (opens) for 0.5 seconds
- Gate stops completely

4. Priority Control

Security panel commands take priority over driver panel commands when both are active simultaneously.

5. FreeRTOS Task Design

Task	Priority	Description
Input Task	High	Reads and debounces all buttons; sends events to a queue
Gate Control Task	Medium	Implements the state machine; decides gate actions
LED Control Task	Medium	Controls Red/Green LEDs based on gate movement direction
Safety Task	Highest	Monitors obstacle signals; enforces safety behavior
Status Task (Optional)	Low	Displays system state using LEDs or UART

6. Inter-Task Communication

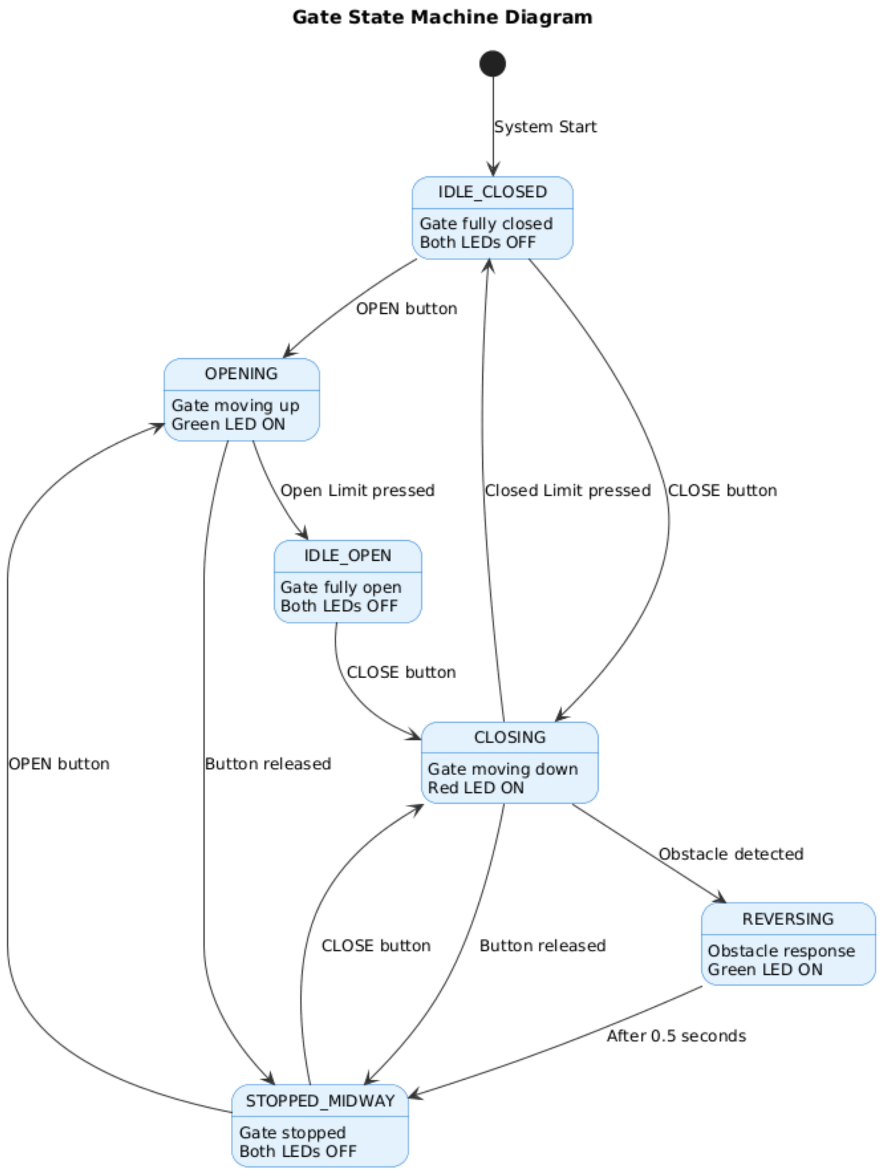
Mechanism	Purpose
Queue	Send button events and commands between tasks
Semaphore	Signal limit button activation
Mutex	Protect shared gate state variable to avoid race conditions

7. Gate State Machine

States

State	Description
IDLE_OPEN	Gate is fully open and stationary
IDLE_CLOSED	Gate is fully closed and stationary
OPENING	Gate is moving up (Green LED ON)
CLOSING	Gate is moving down (Red LED ON)
STOPPED_MIDWAY	Gate stopped between limits
REVERSING	Gate reversing due to obstacle detection

State Diagram



8. Deliverables

- Source Code: Complete FreeRTOS project with all tasks implemented
- Demo Video: Recorded demonstration showing all test cases with explanations
- Documentation Report: System design, state machine diagrams, and test case results

9. Grading Criteria

Marks	Activity
2	Manual Mode works correctly (gate moves while button held, stops on release)
2	One-Touch Auto Mode works correctly (gate moves to limit)
2	Obstacle detection triggers reverse behavior (stops, reverses 0.5s, stops)
2	Limit buttons stop gate movement correctly (open limit and closed limit)
1	Security panel has priority over driver panel when both active
1	Conflicting inputs (OPEN + CLOSE) result in safe stop
2	Tasks are created and scheduled correctly
2	Task priorities are assigned properly
2	Semaphores/mutexes used for resource management
2	Queues used for passing data between tasks
2	Full Technical Report
20	**Total**

10.Test Case Scenarios

Performing these test cases is essential for demonstrating system functionality and acquiring full marks according to the grading criteria. Each test case is mapped to specific grading requirements, and successful execution during the demo video will validate the implementation.

1. **Manual Mode Tests:**
   - TC-01, the driver/security presses and holds the OPEN button while observing the Green LED turning ON; upon releasing the button before reaching the open limit, the gate should stop and transition to STOPPED_MIDWAY state.
   - TC-02 follows the same procedure for closing, where holding the CLOSE button activates the Red LED, and releasing it stops the gate mid-position.
   - TC-03 verifies that manual control works correctly from the STOPPED_MIDWAY state in both directions.
2. **One-Touch Auto Mode Tests:**
   - TC-04 tests automatic opening by briefly tapping the OPEN button from IDLE_CLOSED state; the Green LED should remain ON after release until the Open Limit button is pressed, transitioning to IDLE_OPEN.
   - TC-05 tests the reverse scenario for automatic closing.
   - TC-06 confirms that auto mode functions correctly when initiated from the STOPPED_MIDWAY state.
3. **Obstacle Detection Tests:**
   - TC-07 is the primary safety test where, during auto-closing with the Red LED ON, pressing the Obstacle button should immediately stop the gate, reverse it for 0.5 seconds with the Green LED ON, then stop completely in STOPPED_MIDWAY state.
   - TC-08 verifies this safety behavior overrides even a held CLOSE button.
   - TC-09 confirms that obstacle detection is correctly ignored during opening operations.
4. **Limit Button Tests:**
   - TC-10 and TC-11 verify that pressing the Open Limit button during opening, or the Closed Limit button during closing, immediately stops the gate and transitions to IDLE_OPEN or IDLE_CLOSED respectively.
   - TC-12 ensures that pressing the wrong limit button (e.g., Closed Limit while opening) has no effect.
5. **Security Priority Tests:**
   - TC-13 tests that when both Driver CLOSE and Security OPEN are pressed simultaneously, the security command takes priority and the gate opens.
   - TC-14 tests the reverse scenario where Security CLOSE overrides Driver OPEN.
6. **Conflicting Input Tests:**
   - TC-15 and TC-16 verify that when OPEN and CLOSE buttons are pressed simultaneously on the same panel, whether from a stopped or moving state, the gate stops safely.
7. **RTOS Mechanism Tests:**
   - TC-19 verifies that the queue correctly delivers all button events without loss under rapid input.
   - TC-20 confirms that the mutex properly protects the shared gate state variable from race conditions.
   - TC-21 validates that semaphores correctly signal limit button activation for immediate gate stopping.